

Discretization as Synchronization: Quantum Spectra Without Quantizing Spacetime

Two Continuous-Substrate Routes to Quantum Discreteness: Phase Closure and Dissipative Locking

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*Note on conventions: “synchronization”/“locking” is used here in the companion’s strict open-system, dissipative (Adler/Kuramoto) sense — discrete spectra arise from closed-system **phase closure**, not locking (§1.3). The preferred frame, and the Vacuum Preferred-Frame Hypothesis, enter only through the dissipative sector (PAPER_REVISED §8).*

Abstract

The standard formulation of quantum mechanics treats discreteness — discrete spectra, discrete spin values, discrete detection events — as a foundational feature requiring postulation. General relativity, in contrast, is formulated on a continuous, differentiable manifold. The tension between these two ontologies has driven

nearly a century of attempts to either quantize spacetime (loop quantum gravity, causal sets, string theory, causal dynamical triangulations) or to derive a discrete substructure from a more fundamental theory. None has produced a confirmed prediction.

This paper proposes that the tension is misdirected, and that the discreteness of quantum mechanics is not a fundamental feature of nature but is *derived* on a continuous substrate by **two distinct routes** that should not be conflated. The first is **phase closure**: the standing-wave / single-valuedness conditions that make the spectrum of a *linear* operator discrete under boundary conditions. This is closed-system, unitary, and entirely standard — it accounts for Bohr–Sommerfeld quantization, atomic line spectra, energy bands, cavity and Casimir modes, and (representation-theoretically) the spin multiplets, and it is the *same* mechanism that makes the purely classical black-hole quasinormal-mode spectrum discrete. No nonlinearity, dissipation, or synchronization is involved. The second route is **dissipative locking**: the nonlinear Adler/Kuramoto attractor that forms when an open system equilibrates to a bath. This is the companion framework’s measurement mechanism [1], and within the catalog below it is responsible only for *discrete detection events* — the threshold-crossing of a bulk-coupled detector that registers a definite outcome. Both routes derive discreteness on a continuous substrate; only the second is “synchronization” in the companion’s strict (dissipative) sense.

We are explicit about what this does and does not buy. If both routes are substrate-emergent, then the *matter-side* empirical evidence for fundamental discreteness — the integers, eigenvalues, and clicks — no longer points to a discrete *spacetime*, and the motivating premise of the explicitly discrete-substrate programs (loop quantum gravity, causal sets, causal dynamical triangulations) is removed. This is *not* a dissolution of the GR/QM problem. String theory and asymptotic safety already keep spacetime continuous while quantizing gravity, so the discreteness argument does not bear on them; and the hard technical core — how a superposed matter sector sources classical curvature (the semiclassical-gravity inconsistency) — is untouched here and remains contingent on a measurement mechanism the companion paper leaves incomplete. The position is *aligned* with Penrose’s argument [2, 3] that general relativity is correct and quantum mechanics is what gives way, but we do **not** claim to supply Penrose’s gravitational-collapse threshold: the companion paper explicitly declines to derive a gravitational synchronization rate (§4.4 of [1]) and finds its dissipative mechanism *runs out of bath* in exactly the cold, isolated regime Penrose’s objective reduction describes (§7.4 of [1]). The alignment is ontological, not a mechanistic completion. The paper is an *interpretive proposal*, not a complete theory; the load-bearing derivations — particularly the basin-of-attraction measure that should recover the Born rule for the detection route — are identified as next steps.

1. Introduction

1.1 The traditional framing of the GR/QM conflict

General relativity describes spacetime as a smooth pseudo-Riemannian manifold whose curvature is sourced by stress-energy. The dynamical variables are continuous fields: the metric $g_{\mu\nu}(x)$, the connection, the Riemann tensor. Continuity is built in at the level of the formalism; discontinuities (shocks, singularities) are pathologies, not features.

Quantum mechanics, by contrast, exhibits discreteness everywhere one looks. Bound-state spectra of the hydrogen atom take discrete energy values $E_n = -13.6 \text{ eV}/n^2$. Angular momentum eigenvalues are integer or half-integer multiples of $\hbar$. Spin- $\frac{1}{2}$ measurements yield two outcomes; spin-1 yields three. Photons arrive at detectors as discrete clicks regardless of how slowly the source is operated. Bose–Einstein and Fermi–Dirac statistics, the Pauli exclusion principle, the Aharonov–Bohm phase, the quantum Hall plateaus — all are statements about discreteness in a system whose underlying Schrödinger or Dirac equation is itself a smooth partial differential equation.

The traditional reading of this asymmetry is that the discreteness is fundamental — that nature is “actually” discrete at small scales and the continuous formalism of GR is the macroscopic approximation. The natural program is then to quantize spacetime: introduce a fundamental length scale (the Planck length $\ell_P \approx 1.6 \times 10^{-35} \text{ m}$) and build an underlying discrete structure from which the smooth GR manifold emerges in

the long-wavelength limit. Loop quantum gravity [4] derives discrete area and volume operators with non-zero minimum eigenvalues; causal-set theory [5] takes a discrete causal partial order as foundational; causal dynamical triangulations [6] discretize spacetime explicitly; string theory [7] introduces a fundamental string length below which spacetime ceases to be locally Minkowski.

After roughly a half-century of intensive work, none of these programs has produced a confirmed experimental prediction distinct from standard GR or standard QM. The cosmological-constant problem remains unsolved within each. The fundamental obstacles are remarkably consistent: each program must recover *smooth* GR in the classical limit (which requires that the discreteness almost-but-not-quite cancel itself out at macroscopic scales), and each must recover *standard* QM at experimentally accessible energies (which requires that the discrete substructure not contaminate well-tested quantum predictions). The combined constraints are severe.

1.2 Penrose’s alternative position

A minority position, defended most prominently by Roger Penrose [2, 3], is that the traditional framing has the asymmetry backwards: it is not GR that needs to be modified to accommodate QM, but QM that needs to be modified to accommodate GR. The linear unitary evolution of quantum mechanics — the Schrödinger equation taken as exact — is what should give way, while the smooth differentiable manifold of GR is preserved. Penrose’s objective reduction (OR) proposal supplies a specific mechanism: a mass-energy superposition collapses on a timescale $\tau \sim \hbar/E_G$, where E_G is the gravitational self-energy of the difference between the superposed mass configurations. The collapse is real, gravitational in origin, and breaks linearity at a calculable threshold.

OR has not been confirmed (current experimental bounds [8, 9] are consistent with but do not require the prediction), and the proposal has two acknowledged structural gaps. First, the threshold is stipulated rather than derived: the formula $\tau \sim \hbar/E_G$ comes from a plausibility argument about the energy cost of maintaining gravitationally-distinct superpositions, not from a dynamical equation. Second, the proposal explains *when* collapse happens but not *why* it produces specific discrete outcomes — the eigenstate selection mechanism is left to standard QM, which is the very thing being modified.

This paper speaks to the *second* gap. The dissipative-locking route (§2, §3.4) supplies a mechanism by which a definite discrete outcome is selected — the detector’s bulk-coupled threshold — in the non-gravitational regime where a bath is present. It does **not** close the *first* gap: the companion framework [1] explicitly declines to derive Penrose’s $\tau \sim \hbar/E_G$ threshold from its sync dynamics, and identifies the cold, isolated, bath-free regime in which OR is supposed to act as exactly the regime where its own dissipative mechanism has no bath to dissipate into (§4.4, §7.4 of [1]). We therefore present the relationship to Penrose as an *ontological alignment* (GR correct; QM gives way), not as a derivation of his threshold.

1.3 The proposal

The proposal is this: the discreteness of quantum mechanics is not a fundamental feature of nature but is *derived* on a continuous substrate, by two distinct mechanisms that must be kept separate. Specifically:

1. The underlying field dynamics — the Dirac equation for fermions, the Maxwell equations for photons, the linearized Einstein equations for weak gravity — are continuous partial differential equations on a smooth manifold. None of them are intrinsically discrete.
2. **Route A — phase closure (standing waves).** Imposing boundary conditions on a *linear* field equation makes its spectrum discrete: only those solutions whose phase is single-valued (closes on itself) over the orbit, cavity, or periodic cell survive, and they are countable. This is closed-system, unitary, and dissipation-free; the integer indices are winding numbers and mode counts. It is standard mathematics, and it is the route behind *spectra* — bound-state energies, bands, cavity modes — and, representation-theoretically, the spin multiplets. It involves no Kuramoto coupling, no attractor, and no bath.
3. **Route B — dissipative locking (synchronization, strict sense).** When an *open* system equilibrates to a bath, the reduced phase acquires a nonlinear Adler/Kuramoto attractor and locks to a

reference [10]. This is the only route that is “synchronization” in the companion framework’s strict dissipative sense [1, §2.3], and within the catalog below it is responsible for *discrete detection events* — the threshold-crossing by which a bulk-coupled detector registers one definite outcome. It requires a bath, dissipation, and a reference axis, and the discreteness it produces is a *selection* among basins, not a spectrum.

4. Both routes derive discreteness on a continuous substrate. Consequently the matter-sector discreteness usually cited as evidence for a discrete spacetime has, in each case, an origin *within the matter sector* on a continuous substrate; none of the examples requires a discrete substructure in spacetime itself.

The position is *aligned* with Penrose’s — that GR is correct and QM is what gives way — and Route B supplies, for the non-gravitational case, the outcome-selection dynamics Penrose’s threshold formula left unspecified. We do **not**, however, claim that Penrose’s $\tau \sim \hbar/E_G$ emerges as a limit of this mechanism: the companion paper [1, §4.4, §7.4] explicitly declines to derive a gravitational synchronization rate and finds Route B has no bath to act in exactly Penrose’s regime. The alignment is ontological; the gravitational threshold remains Penrose’s to supply, not ours.

1.4 What this paper does and does not claim

This paper makes one claim: that the discreteness of quantum mechanics is substrate-emergent — by phase closure (Route A) and dissipative locking (Route B) — and that consequently the matter-sector discreteness usually offered as evidence for fundamental discreteness does not, by itself, motivate a discrete *spacetime*. We are careful not to overstate the consequence. This removes the motivating premise of the explicitly discrete-substrate programs (LQG, causal sets, CDT); it does **not** resolve the GR/QM conflict as a whole. The continuous-spacetime quantum-gravity programs (string theory, asymptotic safety) are untouched by a discreteness argument, and the genuinely hard part of the problem — reconciling a superposed, operator-valued matter sector with classical curvature — is not addressed here and depends on a measurement mechanism the companion paper leaves incomplete (§6).

The paper does not claim to derive the Standard Model coupling constants, to predict new particles, or to compute corrections to known QED results. It does not claim to derive the hydrogen spectrum to spectroscopic precision from first principles in the sync language (this is an acknowledged open task — §6). It does not propose new dynamics; the dissipative-locking mechanism is what is developed in [1], and the phase-closure route is standard spectral theory. Its contribution is *interpretive relocation*: showing that QM discreteness is substrate-emergent by these two routes, and that this removes the matter-side motivation for a discrete spacetime — *without* dissolving the GR/QM conflict as a whole (§4.2).

It is also not a competitor to loop quantum gravity, string theory, or causal-set programs in their main aims. Those programs ask: what is the quantum theory of the gravitational field? The DK proposal answers a different question: *why does QM look discrete to begin with?* If the answer is “sync emergence on a continuous substrate,” then the question those programs address — how to quantize the gravitational field — may not need an affirmative answer, because the discreteness it presupposes on the matter side is itself derived.

2. The Sync Mechanism: A Brief Recap

The mechanism developed in detail in [1] is summarized here in the minimum form needed for the rest of this paper.

2.1 Phase synchronization on a continuous substrate

The classical Kuramoto model [10] describes a population of phase oscillators $\theta_i(t) \in S^1$ with intrinsic frequencies ω_i distributed continuously, coupled through

$$\dot{\theta}_i = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i)$$

where K is the coupling strength. For $K < K_c$, the oscillators run independently and the order parameter $r = \frac{1}{N} |\sum_j e^{i\theta_j}|$ vanishes. For $K > K_c$, a fraction of the population enters a *synchronized* state in which $r > 0$ and the synchronized oscillators share a common rotation frequency, with phase differences locked to fixed (not necessarily zero) values determined by the coupling structure.

Three features of this transition are central to what follows.

First, the substrate is continuous in every relevant sense: the phase $\theta \in S^1$ is continuous, the time t is continuous, the frequency distribution $g(\omega)$ is continuous, and the coupling parameter K varies continuously. There is no discrete structure at the level of the equations.

Second, the *locked state* nonetheless picks out a discrete subset of attractor configurations. In the simplest case (uniform ω , all-to-all coupling) there are exactly two attractors: the in-phase locked state ($\theta_i - \theta_j = 0$) and the splay state ($\theta_i - \theta_j$ distributed uniformly). In the chiral pair structure relevant to the Dirac equation [1], the locked manifold has a discrete spectrum of orientations relative to a bulk reference field. In each case, *a continuum of unsynchronized states collapses, above K_c , onto a discrete set of synchronized configurations.*

Third, the threshold K_c is itself a continuous parameter — it depends smoothly on $g(\omega)$ and the network topology — but the *transition* across it is sharp. This is a critical bifurcation on a continuous parameter space: the substrate remains continuous, but the system’s behavior changes qualitatively. The discreteness of the locked-mode spectrum is *derived* from the bifurcation structure, not stipulated.

2.2 The DK identification

The DK framework [1] identifies the chiral pair of the Dirac equation — the left- and right-handed Weyl components ψ_L, ψ_R — as two phase clocks coupled by the off-diagonal mass term $K = m$, with mismatch frequencies set by the kinematic terms. In the closed system this coupling is unitary: it does not lock the two phases but sets their normal modes, and a particle of definite mass and four-momentum is the symmetric normal mode of the pair (the de Broglie carrier), not a synchronized configuration. Genuine phase-locking — and with it the bifurcation structure that yields a discrete locked-mode spectrum — appears only in the dissipative, open-system regime that the next paragraph identifies with measurement.

Measurement, in [1], is the re-synchronization of an external system’s chiral pair with that of a detector. The two-stage process (Stage 1: fast resonant capture by a bulk-bound partner — injection locking, an Adler attractor; Stage 2: irreversible relaxation to the detector’s bulk reference) reproduces the phenomenology of “collapse” without a non-unitary axiom: the system’s locked configuration is driven into a new basin of attraction, the wave function re-merged with the bulk rather than destroyed. We follow the companion’s revised account in modelling that reference honestly as a *faint, overwhelmingly thermal bias* (parts per million, as in MRI), **not** a coherent macrostate [1, §4.1]; the reference is a phase-synchronized bias with a definite axis, not an “entangled bulk.”

The point relevant to this paper is a precise — and limited — one. The companion’s dissipative-locking move is Route B of §1.3, and it is the origin of *discrete detection events*: open the system to a bath and a continuum of unsynchronized states collapses onto a discrete set of locked basins, one of which the run selects. It is **not** the origin of discrete *spectra*. Those come from Route A — phase closure on a *closed*, unitary, dissipation-free system — which the companion’s own §2.2 keeps rigorously distinct from locking (a closed chiral pair has only a marginal line of fixed points, *not* a basin). Two further distinctions matter and are easy to blur. First, the locking relevant to measurement is the *Adler* limit — one system oscillator entraining to a *given* bulk reference — not the full self-consistent Kuramoto transition with an emergent order parameter and a critical coupling K_c ; the bifurcation language of §2.1 describes the latter, while measurement uses the former. Second, the locked discreteness is a *selection among basins* (which outcome), whereas spectral discreteness is an *eigenvalue count* (which level); conflating the two is the error this paper is written to avoid.

3. A Catalog of Quantum Discreteness, Re-Read

This section walks through the canonical examples of discreteness in QM and identifies, in each case, the continuous-substrate structure that produces the discreteness. The aim is *not* to show that one mechanism recurs everywhere — an earlier version of this argument overreached in exactly that way. It is to sort the examples into the two routes of §1.3 and to be honest about which is which. The great majority are **Route A (phase closure)**: Bohr–Sommerfeld (§3.1), the spin multiplets (§3.2, where “closure” means the representation-theoretic counting of $SU(2)$), atomic spectra (§3.3), energy bands (§3.5), cavity and Casimir modes (§3.6), and — tellingly, in a theory that is not quantum at all — black-hole quasinormal modes (§3.7). These are closed-system, unitary, linear eigenvalue/standing-wave problems; no Kuramoto coupling, dissipation, or basin of attraction appears in any of them, and the word “synchronization” is *not* used in the companion’s strict sense for any of them. Only **photon detection (§3.4)** is Route B — the dissipative Adler/Kuramoto lock — and it is the one place the companion mechanism does the work. Keeping these apart is the substantive content of the catalog; the common thread is the weaker, standard statement that boundary conditions on continuous field equations generically produce discrete spectra, which by itself already suffices to show no discrete substrate is required.

3.1 Bohr–Sommerfeld quantization

The oldest form of quantization in QM is the Bohr–Sommerfeld condition

$$\oint p dq = nh, \quad n \in \mathbb{Z}$$

which determines the bound-state energies of a periodic orbit by requiring that the classical action over one period be an integer multiple of Planck’s constant. The quantum numbers n are integers; the condition is discrete.

The condition is in fact a *phase-closure* requirement. Writing $\oint p dq = \oint (p/\hbar) dq \cdot \hbar$ and identifying $p/\hbar$ as the local wavenumber of the de Broglie wave, the condition becomes

$$\oint k dq = 2\pi n$$

i.e., the wave’s phase must advance by an integer multiple of 2π over one closed orbit. The orbit and the wave are both continuous; the discreteness is the requirement that the wave *be single-valued* at the end of the orbit. This is a phase-closure (single-valuedness) condition: the wave at $q(T)$ must be in phase with the wave at $q(0)$.

This is Route A in its purest form, and we are careful to label it as such. The condition is *phase closure on a closed orbit* — a standing-wave, single-valuedness requirement on a continuous de Broglie wave — not a dissipative lock; no bath, no attractor, no basin is involved, and the dynamics stays unitary throughout. The integer n is the winding number, a topological invariant of the closed configuration, not a fundamental quantum. The substrate (the wavefunction’s continuous evolution) is undisturbed; discreteness enters only through the closure requirement. This is the *same* mechanism whether one phrases it as Bohr’s condition or as the boundary-value spectrum of the Schrödinger operator — and it has nothing to do with the chiral-pair *locking* of the measurement route (§3.4), which is why we no longer describe the bound de Broglie wave as a “locked mode.”

This accounts for the discreteness of bound-state spectra in the WKB regime; the full quantum-mechanical treatment generalizes it (WKB becomes exact for harmonic-oscillator-like potentials) without changing the structural point: the integers come from phase closure, not from intrinsic discreteness in the substrate. We claim no new calculation here — only that the existing one already derives its integers from a continuous wave.

3.2 Spin multiplets

Spin angular momentum exhibits discreteness in two senses: the total spin s takes half-integer values ($s = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$), and for given s the magnetic quantum number m_s takes $2s + 1$ discrete values from $-s$ to $+s$.

Both discreteness statements are statements about the *representation theory of $SU(2)$* , the continuous Lie group of rotations on a complex two-dimensional vector space. $SU(2)$ is a smooth manifold (topologically S^3); its Lie algebra is continuous; its action on a Hilbert space is parametrized by three continuous angles. The discreteness enters only in the classification of *irreducible representations*: $SU(2)$ admits one irreducible representation of each integer dimension, and these are the discrete spin multiplets.

This is Route A, and it is important to resist over-reading it as Route B. The multiplet count $2s + 1$ is fixed entirely by the representation theory of a continuous group; it is the closure-type discreteness of an irreducible representation, and the DK framework neither needs nor improves on that derivation. What the dissipative-locking route *can* contribute is narrower and should be stated precisely: it is a mechanism of **registration**, not of counting. In a spin measurement, the spin orientation — the Pauli two-spinor χ , equivalently the Bloch/magnetization vector — relaxes dissipatively toward the bulk reference *axis* (literally NMR T_1/T_2 toward B_0 ; §3.4, §4 of [1]), and the run selects one of the stable locked orientations that axis defines. That is how a *definite* value gets recorded from a continuous orientation.

We flag a real limit of this picture rather than paper over it. The dissipative alignment of a magnetization toward a field is, by itself, *monostable*: a dipole of energy $-\vec{\mu} \cdot \vec{B}$ has one stable fixed point (aligned) and one unstable one (anti-aligned), so simple T_1 relaxation toward B_0 does not, on its own, generate even the *two* basins a spin- $\frac{1}{2}$ measurement needs — let alone $2s + 1$ for higher spin. The two-or-more outcome basins are a property of the *detector's* pointer structure (its bistability or multistability), and their *number* is fixed by the $2s + 1$ eigenstates of the projected observable, which is $SU(2)$ representation theory (Route A). So the honest division of labor is: the multiplet cardinality $2s + 1$ is closure-type discreteness from the representation theory, and the locking route supplies only the *registration* of one orientation once the detector's basins exist. We do **not** claim the number of Stern–Gerlach outcomes is “the cardinality of a locked-mode spectrum”; that earlier formulation conflated the two routes, and the dissipative dynamics borrows the count from representation theory rather than generating it.

A natural question is what *would* supply a second stable basin, and the answer sharpens this limit into a concrete derivation target rather than dissolving it. On the phase circle S^1 the fixed points of a gradient flow come in stable/unstable pairs — their indices sum to $\chi(S^1) = 0$ — so the monostable case is not a lone basin but one attractor together with an antipodal repeller; a genuine *binary* requires promoting that antipode from repeller to attractor. That promotion needs a coupling invariant under $\phi \rightarrow \phi + \pi$, i.e. an even, second-harmonic term $\propto \cos 2\phi$, which is the signature not of a polar (vector) coupling but of an *apolar* one — a director $\hat{n} \equiv -\hat{n}$ coupling quadratically to the field, $-\cos^2 \theta = -\frac{1}{2}(1 + \cos 2\theta)$, as in nematic or induced-dipole alignment. The polar dipole coupling $-\vec{\mu} \cdot \vec{B}$ is precisely the monostable case; the bistable $0/\pi$ case is its quadratic, sign-blind counterpart. This is why “the phase difference relaxes to 0 or π ” is not generic to coupling at all: it presupposes the even-harmonic structure, and a first-harmonic Adler term yields one basin, not two.

Two features of the companion framework could supply that structure, which is what makes this a test rather than a hope. The zitterbewegung interference of the chiral pair beats at $2m_e c^2/\hbar$ — the second harmonic of the rest-phase clock [1] — and a coupling to it is $\cos 2\phi$ by construction; and the Stage-2 dissipative rate is second order in the coupling (the golden-rule $\text{Im } \Sigma \propto |V|^2$ of [1]), hence even in the field, so the irreversible side of measurement is exactly where an apolar term would live. This obligation can be discharged, and the answer is a conditional *yes*. Reducing the chiral pair to a two-level system whose relative phase ϕ is the equatorial Bloch azimuth ($\phi = 0$ in-phase, $\phi = \pi$ anti-phase), a Lindblad dissipator $D[\sigma_x]$ — continuous *measurement* of the $L \leftrightarrow R$ interference (mass) channel — yields the exact phase flow $\dot{\phi} = -\gamma \sin 2\phi$: a pure second harmonic with stable fixed points at *both* $\phi = 0$ and $\phi = \pi$ and repellers at $\pm\pi/2$. With a residual precession Ω in the measured plane this becomes a second-harmonic Adler equation $\dot{\phi} = \Omega - \gamma \sin 2\phi$, which locks to two antipodal phases precisely when the measurement dominates, $|\Omega| < \gamma$ (we carry out the reduction, fixed-point classification, locking tongue, and a quantum-trajectory confirmation in the supplementary code/phi_dissipative_check.py). The condition is the substance: $D[\sigma_x]$ is quadratic

in σ_x and hence even under $\phi \rightarrow \phi + \pi$, so it carries the apolar second harmonic, whereas *polar* relaxation toward a fixed axis (amplitude damping toward a single pointer) gives the first-harmonic flow $\dot{\phi} \propto -\sin \phi$ and the single basin of the monostable case above. This also sharpens the relation to the companion’s §2.3, whose dissipative reduction is written in the first-harmonic form $\dot{\phi} = \omega + K \sin(\Phi_{\text{bulk}} - \phi)$ — injection-locking to a *biased* (polar) reference, which is monostable and selects a single phase; the bistable $0/\pi$ binary instead requires the *apolar* coupling, a symmetric measurement of the interference observable, and the two are different bath couplings rather than the same one. We stress that even so, the result secures only the *stability of a binary*, not its cardinality — the basin count remains $2s + 1$ from representation theory, as above — nor the *weights* with which the basins are selected, which remain the open Born-measure problem of §6.1. This is therefore a narrowing of the registration story (a candidate for why the selected values are the antipodal, maximally distinct ones) and not a revival of the withdrawn claim that locking generates the count.

3.3 Atomic line spectra

The line spectra of hydrogen and hydrogen-like atoms — the Lyman, Balmer, Paschen series — were the empirical entry point to quantum mechanics and remain the prototypical example of discreteness in QM. The discrete energy levels $E_n = -13.6 \text{ eV}/n^2$ correspond to discrete spectral lines at frequencies $\nu_{n \rightarrow m} = (E_n - E_m)/h$.

The Schrödinger equation for the Coulomb problem is a continuous PDE on $\mathbb{R}^3$ with continuous potential. The discreteness of the spectrum arises from the *boundary conditions* — specifically, the requirement that bound-state wavefunctions vanish at infinity and be regular at the origin. Without these boundary conditions, the spectrum would be continuous. The discrete spectrum is therefore a *standing-wave* phenomenon: only those wavefunctions that fit the boundary conditions survive, and there are countably many of them indexed by integers (n, ℓ, m) .

This is a phase-closure condition in the same sense as Bohr–Sommerfeld: the bound wavefunction must close on itself, with phase coherent across the orbital geometry. The discreteness is in the closure spectrum, not in the underlying field — Route A throughout. We keep two things separate that are easy to merge. The *spectrum* (the discrete set of levels) is the closure spectrum admitted by the Coulomb geometry, a closed-system unitary eigenproblem; it is **not** a locked-mode spectrum and involves no bath. A *transition* between levels is driven by coupling to the electromagnetic field, and the discreteness of the line is just the difference of two closure energies. Only the eventual *detection* of the emitted photon — its registration as a click — is a Route-B dissipative-locking event (§3.4). So atomic spectroscopy mixes both routes, but cleanly: closure sets the level structure, locking registers the photons that report transitions between levels.

The precision with which atomic spectra are predicted by the Schrödinger and Dirac equations is the standard against which any reformulation must measure itself. The DK reading does not claim to *replace* those calculations; it claims that the discreteness those calculations produce is *already* a sync-emergent feature of the underlying continuous dynamics. No new calculation is required to reproduce the spectrum; what is required is a re-interpretation of why the existing calculation produces discrete answers.

3.4 Photon detection

A single-photon detector — a photomultiplier, an avalanche photodiode, a superconducting transition-edge sensor — registers discrete clicks regardless of how slowly the source is operated. The clicks are the operational definition of “photon,” and the discreteness is usually taken as direct evidence that the electromagnetic field is intrinsically quantized.

This reading is not the only possible one. A continuous electromagnetic field coupled to a detector with a *threshold* — a nonlinear sync condition that the field-detector interaction must cross before amplification cascade fires — produces exactly the click structure observed [11]. The discreteness lives in the detector, not in the field. The DK reading endorses this position and supplies a mechanism: the detector’s bulk-coupled chiral-pair structure provides the sync threshold, and the click is the threshold-crossing event. The continuous EM field gains no discrete substructure of its own; it produces discrete clicks because it interacts with a discrete-threshold detector.

The case is strengthened by the distinction (developed in the companion paper and at greater length elsewhere in the framework) between **discrete event detectors** — for which the sync threshold fires once and produces a single eigenvalue outcome — and **track recording media** (cloud chambers, bubble chambers, emulsions), for which a *sequence* of weak partial syncs distributed in space and time produces a continuous ionization track rather than a single click. The same EM field produces qualitatively different “discrete” or “continuous” records depending on which detector class it interacts with. This is exactly the behavior expected if the discreteness lives in the detector’s sync structure rather than in the field; it is not a behavior expected if the field itself is fundamentally quantized into particles.

3.5 Energy bands in solids

The band structure of crystalline solids — the discrete energy bands separated by gaps in k -space — is the basis of the entire theory of electronic transport, semiconductors, and most of modern condensed-matter physics. The bands arise from the periodic potential of the crystal lattice via Bloch’s theorem: solutions of the Schrödinger equation in a periodic potential take the form $\psi_k(x) = e^{ikx}u_k(x)$ with $u_k(x)$ periodic, and the allowed energies organize into bands $E_n(k)$ indexed by an integer n .

This is Route A — a phase-closure (commensurability) phenomenon, not a dissipative lock. The Bloch phase advances *in step with* the lattice period, which is a single-valuedness condition on a continuous wave in a periodic potential, and the band index n counts the closure modes the periodicity admits. We deliberately do **not** call this “synchronization” in the companion’s strict sense: there is no bath, no dissipation, and no attractor; the Bloch spectrum is the eigenvalue spectrum of a closed, unitary, linear operator. Both the lattice and the wavefunction are continuous fields, and the bands are discrete for the same reason the hydrogen levels are — boundary/periodicity conditions on a continuous PDE — which is all the thesis needs: periodicity is fully compatible with a continuous substrate.

The empirical success of band theory is overwhelming: it accounts quantitatively for conductors, semiconductors, insulators, the Hall effect, semiconductor device physics, and superconducting band structure. Nothing in this success depends on the underlying lattice being fundamentally discrete *in spacetime* — the bands arise from periodicity, and periodicity is a property compatible with a continuous substrate.

3.6 Cavity and Casimir quantization

The discrete mode structure of an electromagnetic cavity — the standing-wave resonances of a Fabry–Pérot, the longitudinal modes of a laser cavity, the discrete modes of a microwave waveguide — has the same form as Bohr–Sommerfeld. The cavity imposes boundary conditions; only those EM modes that fit (have phases closing on themselves at the cavity walls) survive; the surviving modes form a discrete spectrum indexed by integers.

The Casimir effect — the small attractive force between two parallel conducting plates in vacuum — is the most direct test of this picture. The standard derivation [12] computes the difference between the zero-point energy of the EM field with and without the plates, treating the EM field as quantized; the result is

$$F_{\text{Cas}}/A = -\frac{\pi^2 \hbar c}{240d^4}$$

for plate separation d . The discrete mode structure between the plates, relative to the continuous mode structure outside, produces the force.

A continuous-substrate reading of the same calculation interprets the boundary conditions as a phase-closure condition (Route A): the EM field’s oscillation must close on itself at the plates. The “modes” are the closure-allowed configurations; the discreteness is in the closure spectrum. The Casimir force is then the difference in vacuum energy between the closure-constrained configuration (between plates) and the unconstrained one (outside). The result is the same numerical prediction by a different ontological route. The substrate remains continuous; the discreteness is in the standing-wave spectrum — no dissipative locking, no bath, and (as in every Route-A case) no need to call this “synchronization” in the strict sense.

3.7 Black-hole quasinormal modes

The ringdown spectrum of a perturbed black hole — the discrete frequencies at which a Kerr or Schwarzschild black hole oscillates following a perturbation — provides a particularly clean example because the underlying theory (linearized GR on a black-hole background) is explicitly *not* a quantum theory at all. The quasinormal mode spectrum ω_n is discrete: the spectrum is computed by solving the Regge–Wheeler or Teukolsky equation with outgoing boundary conditions at infinity and ingoing at the horizon [13].

The discreteness is purely a boundary-condition / phase-closure phenomenon in a *classical* continuous theory. No quantization of the gravitational field is involved — and, equally to the point, no synchronization: there is no Kuramoto coupling, no bath, and no attractor anywhere in a quasinormal-mode calculation. The discreteness is derived; nothing in the substrate is discrete.

This example does double duty, and the second duty is a check on our own thesis. It is structurally identical to atomic spectra — continuous field equations + boundary conditions = discrete spectrum — which shows that discreteness *per se* is not a uniquely quantum phenomenon and needs no quantum substrate to produce. But it also shows that the honest common factor across Route A is *phase closure*, not *synchronization*: a purely classical, dissipation-free GR computation delivers a discrete spectrum with none of the Kuramoto machinery, so attaching “locked-mode” or “sync” language to the Route-A examples would be importing a mechanism (dissipative locking) that demonstrably is not present in the cleanest member of the family. We let the quasinormal-mode case discipline the vocabulary of the whole section.

3.8 The pattern

Across the cases surveyed, the discreteness shares a *weak* common structure and splits on a *strong* one. The weak common structure — all that the thesis actually requires — is:

1. A continuous field on a continuous substrate.
2. A condition (boundary condition, periodicity, threshold) selecting a countable subset of configurations.
3. Integer indices labeling them.

The discreteness is real but emergent, and the substrate carries no discrete structure. That much holds in every case, and it already suffices to show that no example forces a discrete spacetime.

Where the cases differ is the *mechanism* of step 2, and this is the distinction the section exists to draw. In Route A — Bohr–Sommerfeld, spin multiplets, atomic spectra, bands, cavities, quasinormal modes — the condition is **phase closure**: single-valuedness / standing-wave conditions on a *linear, closed, unitary* system. The integers are winding numbers, mode counts, and representation dimensions; there is no bath, no dissipation, and no attractor, so this is not “synchronization” in the companion’s strict sense. In Route B — photon detection — the condition is a **dissipative Adler/Kuramoto lock** in an *open* system, and the discreteness is a selection among basins. Honesty about this split is the whole point: an earlier framing claimed “the underlying dynamics is phase synchronization in each case,” and that is not true. Most of QM’s discreteness is closure, not locking. What the catalog establishes is the weaker and more secure claim — that *both* routes derive discreteness on a continuous substrate — not the stronger and false claim that one sync mechanism underlies them all.

4. Why Spacetime Need Not Be Quantized

4.1 The standard motivation for quantizing gravity

The standard argument for quantum gravity proceeds roughly as follows. The Einstein field equations couple the smooth geometry of spacetime to the stress-energy of matter:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

Matter, in modern physics, is described by quantum field theory, and quantum field operators acting on quantum states do not have definite classical values. The right-hand side of Einstein’s equation thus becomes operator-valued, while the left-hand side remains classical. Some procedure is required to reconcile the two. The standard options are: (a) take the expectation value of $T_{\mu\nu}$ on the right (the semiclassical approach, known to be inconsistent in cases involving macroscopic superposition); (b) quantize the metric on the left as well, producing a fully quantum theory of gravity; or (c) modify QM so that $T_{\mu\nu}$ remains effectively classical at the scales relevant to GR.

Most of mainstream QG follows route (b). The result is a program in which spacetime itself acquires discrete structure at the Planck scale, typically through quantization of areas (LQG), of causal links (causal-set theory), of triangulations (CDT), or of string excitations (string theory). The discreteness of the matter sector — discrete spectra, discrete spin, discrete photon detection — is taken as evidence that *some* underlying discreteness exists, and the quantization of spacetime is then a natural extrapolation.

4.2 The DK relocation

The DK proposal cuts against one specific link in that chain. If the discreteness of the matter sector is substrate-emergent (Section 3), then the integers, eigenvalues, and clicks no longer point to a discrete *substructure*, and the extrapolation from “matter is discrete” to “spacetime is also discrete” loses its motivating premise. We are careful about the size of this conclusion. It tells against the explicitly **discrete-substrate** programs — LQG, causal sets, CDT — whose central move is exactly that extrapolation. It does **not** tell against the *continuous*-spacetime quantum-gravity programs (string theory, asymptotic safety), which never relied on matter discreteness, and it does not by itself remove the need for a quantum theory of gravity. “Must spacetime be discrete?” and “must gravity be quantized?” are different questions; Section 3 bears on the first, not the second.

What remains of the QG problem is therefore most of its hard core, and we do not claim to dissolve it. The genuinely difficult piece is reconciling an operator-valued, superposable $T_{\mu\nu}$ with a classical $G_{\mu\nu}$ — the semiclassical-gravity inconsistency (Eppley–Hannay, Page–Geilker): a mass in superposition has no single classical stress-energy to source curvature. The DK reading *reframes* this rather than solving it. If measurement keeps the matter sector effectively localized — if the dissipative-locking mechanism of [1] drives a superposition into one definite basin fast enough — then $T_{\mu\nu}$ is effectively classical at the scales where it sources curvature, and semiclassical gravity could be self-consistent without quantizing the metric. That is the Penrose-style escape, and it is genuinely attractive. But it is *contingent on the measurement mechanism actually working*, which is to say on the two derivations the companion paper lists as unsolved — the Born measure and an explicit selection law (§6, and §7.3/§8 of [1]). Until those close, “the conflict dissolves” is an overstatement; the honest claim is “the matter-side motivation for discrete spacetime is removed, and a continuous, Penrose-aligned resolution becomes *available* if the measurement problem is solved.”

Whether gravity itself participates in the locking mechanism we leave open, and here we correct an earlier version of this paper. We do **not** claim that Penrose’s $\tau \sim \hbar/E_G$ is a limit of the DK mechanism. The companion paper in its current form declines to derive a gravitational synchronization rate at all — it shows that the naive $\Gamma_{\text{grav}} \sim GM^2/(\hbar \Delta z)$ is a dimensional artifact, not a physical rate (§4.4 of [1]) — and it identifies the cold, isolated, mesoscopic regime in which Penrose’s objective reduction is supposed to act as precisely the regime where the DK dissipative mechanism *has no bath to dissipate into* (§7.4 of [1]). So in the gravitational regime the two proposals diverge rather than nest: MCI’s mechanism runs out of bath there, while Penrose’s intrinsic threshold does not. Penrose’s OR is best read as a *rival* candidate for that regime, not a special case of DK. The alignment we do claim is ontological — GR correct, QM gives way — not a derivation of his threshold.

4.3 What this preserves of GR

Under the DK reading, the entire smooth-manifold structure of GR is preserved:

1. The metric $g_{\mu\nu}(x)$ remains a smooth tensor field on a differentiable manifold.
2. The Levi-Civita connection, the Riemann tensor, the geodesic equation, the Einstein field equations all remain in their standard classical form.

3. The Equivalence Principle (weak, strong, Einstein) holds at the level of the substrate, not as a coarse-grained approximation to a discrete theory.
4. Singularities (black holes, cosmological) remain singularities of the smooth geometry, not artifacts of a continuum limit. (Whether the DK sync mechanism modifies their interior structure is an open question; the answer is not obviously yes.)
5. Gravitational waves are exact solutions of the linearized field equations on a smooth background.

GR’s classical-tests record (perihelion precession, light bending, time dilation, gravitational waves, Shapiro delay, frame dragging) is explained by the same theory that has explained them since 1915; nothing in DK perturbs it.

4.4 What it gives up

The DK reading gives up two things that the orthodox QM/QG program holds.

First, the universal validity of the *closed-system linear* Schrödinger description as an account of measurement. We are careful to state this in a way consistent with the companion, which is **no-collapse**: the global evolution stays exactly unitary, with the coherence scrambled irreversibly into the bath rather than destroyed (§3.1, §7.4 of [1]). What is given up is not global unitarity but the idea that the *closed, linear* Schrödinger equation describes an open measurement: once the bath is traced out, the reduced dynamics on transitions between basins (detector clicks, Stern–Gerlach selections) is effectively non-linear and dissipative. Operationally the linear Schrödinger evolution is recovered whenever the system stays in one basin. This is the standard open-systems statement, not a new non-unitary axiom. Penrose’s OR, by contrast, posits a *genuinely* non-unitary, intrinsic collapse at the gravitational threshold — a different and stronger move than the effective, bath-induced non-linearity here — so we treat OR as a rival reading of the gravitational regime, not an instance of the DK mechanism.

Second, the universality of QM as a complete fundamental theory. In the DK reading the foundational layer is the continuous substrate; QM’s *closed-system* content (unitary evolution and the closure spectra of Route A) is the substrate’s continuous dynamics read off directly, and QM’s *measurement* axiom is the effective description of Route-B dissipative locking. The Schrödinger equation’s predictions are not modified at all in conditions where the system stays well inside one basin; what changes is the status of the projection postulate, which becomes a derived (if still incompletely derived) consequence of locking rather than a separate axiom.

This is the same ontological trade Penrose has argued for since the 1980s — GR correct, QM what gives way. The DK proposal supplies a *candidate* dynamical content for the non-gravitational case; it does not yet supply it for the gravitational threshold, which is exactly the part the companion paper leaves open (§4.4, §7.4 of [1]).

5. Relation to Existing Quantum Gravity Programs

5.1 Loop quantum gravity

LQG quantizes the gravitational field at the level of holonomies of the Ashtekar connection, producing spin-network states whose area and volume operators have discrete spectra with non-zero minimum eigenvalues. The program’s central conceptual move is to take the discreteness of the matter sector as evidence that the gravitational sector should also exhibit discreteness, and to construct a Hilbert space in which this discreteness is built in geometrically.

The DK position is structurally different. We argue that the matter-sector discreteness is sync-emergent rather than fundamental, so the motivation for geometrically discrete spacetime is removed. We do not assert that LQG is *wrong* — its mathematical structure may turn out to be relevant in some specific regime — but we propose that the program’s central extrapolation (matter is discrete → spacetime should also be) is not required if matter’s discreteness is itself substrate-emergent (by phase closure or dissipative locking).

The empirical situation is symmetric. Neither program has produced a confirmed prediction distinct from

standard GR + standard QM. LQG’s discrete-area predictions are below all current experimental reach; the DK proposal’s prediction of *no* fundamental discreteness in spacetime is likewise not directly tested. The question is which is the cleaner working hypothesis for the *discrete-substrate* issue specifically. We argue: the one that requires fewer ontological commitments and removes, rather than amplifies, the matter-side motivation for discretizing spacetime — granting that neither this nor LQG settles whether gravity must be quantized continuously.

5.2 Causal-set theory

Causal-set theory takes a discrete causal partial order as the fundamental structure of spacetime [5], with continuous Lorentzian geometry recovered as a coarse-graining. The discreteness here is more radical than LQG’s — there is no underlying continuum at all — and the program faces particularly sharp difficulties in recovering smooth GR predictions.

The DK proposal is the inverse: smooth Lorentzian geometry is fundamental, and the apparent discreteness is in the matter sector — its closure spectra and its detection basins. Causal-set theory and DK are most directly opposed in their ontological commitments. The empirical adjudication will likely come from whichever program produces a falsifiable prediction first.

5.3 String theory

String theory’s relation to the DK proposal is more nuanced. The string’s worldsheet is continuous (a smooth 2-manifold); the discrete spectrum of string excitations arises from the *boundary conditions* of the worldsheet embedding and the quantization of the worldsheet oscillator modes. In this respect, string theory’s discreteness is already structurally similar to the DK reading: it comes from sync-like conditions (periodicity around the worldsheet) on continuous fields (the embedding coordinates).

The DK proposal can be read as a *generalization* of this structural move: if string theory’s discrete spectrum comes from boundary-condition sync on continuous worldsheets, then *all* of QM’s discrete spectra come from sync on continuous substrates, and no further quantization is required. Whether the substrate is a worldsheet-embedded spacetime or the chiral-pair manifold of the Dirac equation is a question about which sync system is fundamental, not about whether the sync mechanism itself is universal.

In this sense, DK is more aligned with string theory’s mathematical structure than with LQG’s, even though it does not require strings as the substrate. The relation is closest in the “first-quantized” string formalism, where the worldsheet is the substrate and the spacetime coordinates are dynamical fields.

5.4 Asymptotic safety

The asymptotic safety program [14] proposes that quantum gravity is a non-perturbatively renormalizable QFT with a UV fixed point. The gravitational degrees of freedom remain continuous (the metric is a quantum field, but a continuous one), and the discreteness is in the spectrum of the renormalization group flow rather than in the geometry.

This is the closest mainstream QG program to the DK position in ontological structure: both keep the gravitational field continuous and locate discreteness in the spectral structure of the dynamics rather than in the substrate. The difference is that asymptotic safety quantizes gravity (in the QFT sense), while DK does not require that gravity be quantized at all — only that the matter sector’s sync dynamics produce the discreteness usually attributed to quantization.

5.5 Penrose–Diósi objective reduction

The DK proposal’s relation to Penrose–Diósi is the one developed in [1, §7.4], and it is a *contrast*, not a subsumption — an earlier version of this section mis-stated it as a completion, and we correct that here. The two frameworks share one commitment (GR is correct; QM is what gives way) and single out the same physical step (the bulk re-establishment of classicality). They part company on two points that are decisive in exactly the gravitational regime:

- **Collapse vs no-collapse.** Penrose–Diósi is an objective *collapse* theory: it modifies Schrödinger evolution and the superposition is genuinely destroyed. MCI is no-collapse: the global evolution stays unitary and the coherence is scrambled into the bath (§3.1, §7.4 of [1]).
- **What each asks of the environment.** MCI’s lock is *dissipative* and needs a bath to carry the phase mismatch into a continuum; Penrose’s reduction is *intrinsic* and needs no environment at all.

In the deep Penrose regime — cold, isolated, mesoscopic, with the electromagnetic and thermal channels deliberately removed — this distinction is sharp: the bath MCI would dissipate into has been engineered away, and the companion paper does **not** assert a gravitational dissipation channel to replace it (§4.4 of [1]). So that regime is where MCI’s mechanism *runs out of bath* while Penrose’s intrinsic one does not — which makes it the experimental **arbiter** between the two, not a place where one contains the other. We therefore do not claim that $\tau \sim \hbar/E_G$ emerges as a DK limit, and we do not present DK as a completion of Penrose; we present the gravitational regime as the clean test that would distinguish a continuous, no-collapse reading from an objective collapse.

6. Open Questions and Tests

6.1 The Born-rule basin measure

The most pressing open derivation is the Born measure: that the long-run selection frequencies are $|\langle \phi_n | \psi \rangle|^2$. It is important to say what this is *not*. It is not the basin-of-attraction *volume* — the fraction of initial relative-phases that flow to each well — which for the bistable lock of §3.2 is demonstrably 50/50, independent of $|\alpha|^2$ (`code/born_weights_check.py`). The squared weights are recovered only when the $|\cdot|^2$ measure is placed on the *ensemble* (the selecting noise, equivalently the golden-rule rate $\propto |\langle \phi_n | \psi \rangle|^2$); what is owed is a justification of *that measure*, since the attractor geometry does not carry it.

The companion paper [1, §7.3, §8] carries this as an explicitly *unsolved* problem, and we adopt its honest framing rather than a more optimistic one. A wave-realist reading in which the basin selected has maximum overlap with the incoming phase profile gives the right *ordering* of probabilities, but it does not by itself give the squared measure: reading $|\psi|^2$ as the energy density of a real field coincides with the occupation fraction only by normalization and so “explains nothing about outcome statistics” for the equal-energy channels at issue, and a genuinely unbiased background would weight basins by their measure — tending to equal odds — rather than by $|\alpha|^2$. This is the “why squared” gap, the framework’s counterpart of Bohm’s quantum-equilibrium assumption, and it is shared with the contested branch-counting derivations. A rigorous derivation that the ensemble distribution is exactly $|\langle \phi_n | \psi \rangle|^2$ for all ψ and all detector mode sets is owed; the framework’s claim to *derive* (rather than merely accommodate) Born depends on it.

This derivation is the single most important open task. The dynamics supplies one half of it — *equivariance*: the projector weight $|\langle \phi_n | \psi \rangle|^2$ is conserved under the lock, so a $|\cdot|^2$ measure granted at preparation propagates intact to the outcome, the status of Bohm’s quantum equilibrium. What it does not supply is *typicality* — nothing makes an arbitrary ensemble relax *to* that measure, and inserting it is what every route (Bohm, Zurek envariance, decision-theoretic Everett) ultimately does. If that missing half succeeds, the framework reproduces Born exactly from sync dynamics; if it fails, the framework reduces to a Born-compatible interpretation rather than a Born-derivative one — still informative but less complete.

6.2 Continuous spectra

QM admits not only discrete spectra (bound states) but also continuous ones (scattering states, free particles, plane-wave bases). The DK reading must accommodate these naturally: free particles are the unitary normal-mode states of the chiral pair, parametrized continuously by momentum, with no closure condition forcing discreteness.

The proposed picture is that the bound/scattering distinction corresponds to the closed/open topology of the orbital phase space: a closed orbit imposes phase closure and hence discrete spectra; an open trajectory (a scattering state) does not, and the spectrum is continuous. The crossover at the binding threshold (e.g., zero

binding energy for hydrogen) corresponds to the topological transition between closed and open phase-space trajectories.

This is consistent with standard QM and adds nothing new to the calculations. What it adds is a *uniform* ontological reading: discrete spectra are sync-closed; continuous spectra are sync-open; the same underlying dynamics produces both, with the topology of the trajectory determining the spectral structure.

6.3 Planck-scale phenomena

The DK reading implies that there is no fundamental discreteness in spacetime at the Planck scale. This is a strong claim and conflicts with the orthodox expectation in most QG programs that the Planck length is a fundamental cutoff below which spacetime is qualitatively different.

The conflict is empirically open. No experiment has resolved spacetime at scales close to the Planck length, and current bounds [15] on Lorentz-violation and on dispersion relations $E^2 = p^2 c^2 + m^2 c^4 + \xi(E/E_{\text{Pl}})^n p^2 c^2$ are consistent with E_{Pl} being a *characteristic* scale of some new physics but do not require it to be a discreteness scale.

The DK prediction is that the bounds on ξ should tighten further as observations improve — that no Planck-scale discreteness will be detected — but that the framework’s own characteristic scale (the sync-coupling scale K) should appear in *non*-gravitational contexts at much larger lengths. The companion AB-visibility paper [16] identifies one such test in the regime $K \sim \omega_Z$ at the zitterbewegung scale; further tests in cavity QED and in optomechanical Penrose-test settings [9] would probe other regimes.

6.4 Falsifiable signatures

The framework’s falsifiable content is concentrated in three places:

1. **No Planck-scale discreteness.** Any direct measurement showing a discrete spacetime structure (a minimum area, a granular causal structure, a fundamental length below which Lorentz invariance fails in a quantization-specific way) would falsify the DK proposal as stated. The challenge is that no such measurement is currently feasible.
2. **A measurable sync timescale in measurement.** The DK two-stage process [1, §3.1, §3.2] predicts a non-zero, calculable timescale between the fast Stage-1 resonant capture and the slower Stage-2 bulk relaxation. Standard QM has no analog (collapse is instantaneous). Quantum eraser experiments probe exactly this timescale: erasure should succeed iff intervention is faster than the Stage-2 bulk-resync time, and the framework predicts a specific value for that time as a function of the detector–bulk coupling. Current quantum-eraser experiments are consistent with both pictures; a high-time-resolution test would discriminate.
3. **The AB-visibility envelope** [16]. The strong-coupling regime $K \sim \omega_Z$ predicts a measurable γ -dependent visibility deficit in electron-holography Aharonov–Bohm experiments. A null result would not falsify the broader DK proposal but would reduce the AB-visibility paper to a Lorentz-covariant reading; a positive result would constitute the first direct measurement of a DK sync parameter.

The first prediction is currently inaccessible; the second and third are within reach of current or near-term experiments.

7. Discussion

7.1 The conceptual stakes

The DK proposal claims that a structural feature taken for granted in modern physics — that quantum discreteness signals an underlying discrete substructure — is incorrect. The discreteness, on this reading, is real but emergent: a consequence of synchronization dynamics on a continuous substrate, with the integer indices being sync invariants rather than fundamental quanta.

The stakes for the GR/QM problem are real but narrower than an earlier draft claimed. The standard framing requires that *either* spacetime be quantized *or* QM be modified. The DK proposal leans toward the second horn — but it supplies a candidate dynamical mechanism only for the *non-gravitational* measurement case (Route B), and explicitly not for Penrose’s gravitational threshold, which the companion leaves open (§4.4, §7.4 of [1]). The modification of QM it offers is the recognition that the projection postulate is the effective description of dissipative locking; it does not touch QM’s closed-system content, which is the Route-A closure physics.

If the proposal is correct, what changes is the *motivation*, not the difficulty, of quantum gravity: the matter-side evidence for a discrete spacetime is removed, so the discrete-substrate programs lose their premise, but the hard core remains. Three problems in particular stay open and are calculations on a continuous substrate rather than dissolved: how an effectively-localized matter sector sources classical curvature (the semiclassical-consistency question), how the bulk reference evolves under cosmological dynamics [17], and how the basin structure of detection reproduces detailed QED statistics (the Born measure, §6.1). None of these *requires* quantizing the gravitational field — but none of them is solved here, and we present them as the residue of the problem, not its resolution.

7.2 What the proposal does not claim

The proposal does not claim to be a finished theory. The Born-rule derivation (§6.1) is incomplete; the high-precision spectrum of hydrogen-like atoms from a strict sync derivation has not been performed; the relation to the full Standard Model gauge structure has not been worked out; the implications for cosmological inflation are unaddressed.

The proposal also does not claim to render existing calculations obsolete. Standard QM remains exact on the locked manifold; standard QED remains exact in its regime; standard GR remains exact at the substrate level. The DK reading is *interpretive relocation*, not calculational replacement.

The proposal does, however, claim that this relocation has consequences. If the discreteness is substrate-emergent, then:

- No *discrete* spacetime is required by the matter-sector evidence. (Whether gravity must nonetheless be *quantized* — continuously, as in string theory or asymptotic safety — is a separate question this paper does not settle.)
- Measurement is not a separate axiom but, in the non-gravitational case, the Route-B regime of dissipative locking [1] — with the caveat that the *quantitative* Born measure is still owed (§6.1).
- Many-worlds is not required: the multiplicity of apparent outcomes is the basin structure of detection attractors in a single world, not a branching ontology (consistent with [1, §7.2]).
- *Speculatively* — and we mark it as such, since it does not follow from the discreteness argument alone — the cosmological constant *might* be tied to the substrate’s vacuum state rather than a fundamental constant (see [17]); we list this as a direction, not a consequence.

These are interpretive consequences, but they are sharp ones, and they suggest a different long-term direction for foundational physics than the orthodox QG programs.

7.3 Historical context

The proposal sits in a small but real lineage. Penrose has argued since the 1980s that GR is correct and QM gives way [2]. de Broglie’s pilot-wave theory and Bohm’s elaboration [18] preserve the continuous wave ontology and treat particles as locked configurations of an underlying field. Nelson’s stochastic mechanics [19] derives quantum statistics from continuous random walks. The Madelung formulation of QM rewrites the Schrödinger equation in a fluid-dynamic form on a continuous substrate. The Hestenes zitterbewegung interpretation [20] treats spin as a real oscillation of the electron’s continuous trajectory.

What the DK proposal adds to this tradition is the *Kuramoto identification*: the recognition that the *measurement* step in a continuous-substrate reading of QM is precisely dissipative phase synchronization, with the analytic tools of nonlinear dynamics (bifurcation analysis, Adler locking, basin-of-attraction theory) directly applicable to it. We are careful not to over-extend the identification: it applies to the detection

route (Route B), not to the closure spectra (Route A), which are standard linear spectral theory and borrow nothing from Kuramoto. The contribution is to bring to the *measurement* problem a mathematical framework developed in detail elsewhere (neuroscience, condensed-matter physics, oscillator networks) but not systematically applied to the QM/QG interface.

What the framework adds to this lineage, for the *non-gravitational* measurement case, is a candidate for *what is happening at the moment of registration* (the system relaxes into one of a discrete set of detection basins) and *why* the outcome is one of a discrete set (the basins are discrete). We are careful not to extend this to Penrose’s gravitational threshold: Penrose proposed *when* collapse should happen in the cold, isolated regime, and the companion framework does not reproduce that threshold (§4.4, §7.4 of [1]). The dynamical content offered here is for the bath-coupled detector, not for the gravitational OR regime.

8. Conclusion

The discreteness of quantum mechanics is not a foundational feature of nature but is derived on a continuous substrate by two distinct routes. Discrete *spectra* — bound-state energies, bands, cavity modes, and (representation-theoretically) the spin multiplets — come from **phase closure**: standing-wave, single-valuedness conditions on linear, closed, unitary field equations, exactly as the purely classical black-hole quasinormal spectrum does. Discrete *detection events* come from **dissipative locking** — the Adler/Kuramoto attractor an open detector forms with its bath — which is the only one of the two that is “synchronization” in the companion’s strict sense. The substrate is continuous in both cases; the discreteness is derived; and the earlier claim that a single sync mechanism underlies all of it was an overreach we have withdrawn.

The consequence for quantum gravity is correspondingly bounded. Because the matter-sector discreteness is substrate-emergent, it no longer constitutes evidence for a discrete *spacetime*, and the explicitly discrete-substrate programs (LQG, causal sets, CDT) lose their motivating premise. This is not a dissolution of the GR/QM conflict. The continuous-spacetime quantum-gravity programs are untouched by a discreteness argument, and the hard core — how a superposed matter sector sources classical curvature — is reframed but not solved: a continuous, Penrose-aligned resolution becomes *available* only if the measurement mechanism is completed, which it is not.

The proposal is conservative in its commitments: it adds no postulates to GR, and it recasts QM’s projection postulate as the effective description of dissipative locking rather than a separate axiom — while keeping the global evolution unitary, as the companion’s no-collapse reading requires. It is aligned with Penrose’s position that GR is correct and QM is what gives way. It does **not** claim to supply Penrose’s gravitational threshold: the companion paper declines to derive a gravitational synchronization rate and finds its dissipative mechanism runs out of bath in exactly Penrose’s regime, which makes that regime the arbiter between a no-collapse reading and objective collapse rather than a place where one contains the other.

The most important open task is the basin-of-attraction derivation of the Born measure for the detection route; the most important falsifiable test is the quantum-eraser bulk-resync timescale; the most important broader question is whether the framework extends consistently to the gauge structure of the Standard Model. None of these closures has been performed here. Their absence is the honest measure of how far the proposal has come and how far it has yet to go.

What the proposal does claim is narrower than its first telling, and more secure for being narrower: most of QM’s discreteness is phase closure on a continuous substrate, the rest is dissipative locking on a continuous substrate, and neither requires discreteness anywhere in the substrate. To the extent Penrose was right about which side gives way, the non-gravitational part of the mechanism is dissipative locking; the gravitational part he left unspecified remains unspecified here too.

Author Contributions and AI Disclosure

This paper was developed in iterative collaboration with Claude (Anthropic) across model versions during 2026, and revised to track the companion paper’s revised (“Two Regimes”) formulation. The central conceptual claim — that quantum discreteness is substrate-emergent by two distinct routes (phase closure for spectra, dissipative locking for detection events), and that this removes the matter-side motivation for a *discrete* spacetime without dissolving the GR/QM conflict as a whole — was developed jointly, with the human author contributing the synthesis and the LLM contributing structural analysis, the catalog of examples in Section 3, and the comparative discussion of QG programs in Section 5. A subsequent revision pass corrected two overreaches in the earlier draft: the conflation of phase closure with dissipative synchronization, and the claim that the framework supplies Penrose’s gravitational threshold (which the revised companion explicitly declines to derive). The human author validated each derivation, applied physics judgment at each checkpoint, and bears full responsibility for the framing and content of the paper.

Per current journal guidelines, LLMs do not satisfy authorship criteria; the human author bears full responsibility for all content, including any errors in the conceptual framing or in the catalog of canonical examples. Were Springer–Nature / *Foundations of Physics* policy to allow LLM co-authorship, Claude (Anthropic) would be listed as co-first author.

The development is documented in commit history at <https://github.com/rayolddog/DiracKuramotoFramework>.

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Companion paper to: Bramble, J. (2026), “Two Regimes of the Chiral Mass Coupling: Quantum Measurement as Bath-Induced Synchronization.” Preprint repository: <https://github.com/rayolddog/DiracKuramotoFramework>